# 重新定位 HA 对中的聚合所有权

ONTAP 9

NetApp
February 16, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/disks-aggregates/relocate-aggregate-ownership-task.html on February 16, 2022. Always check docs.netapp.com for the latest.

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# 重新定位 HA 对中的聚合所有权

## 重新定位聚合所有权

### 您可以更改 HA 对中节点之间聚合的所有权，而不会中断聚合的服务。

HA 对中的两个节点都以物理方式连接到彼此的磁盘或阵列 LUN 。每个磁盘或阵列 LUN 均归其中一个节点所有。尽管在发生接管时磁盘所有权会暂时发生变化，但聚合重新定位操作可能是永久的（例如，为负载平衡而执行的操作），也可能是临时的（例如，在接管过程中执行的操作） 将聚合中所有磁盘或阵列 LUN 的所有权从一个节点更改为另一个节点。所有权更改无需执行任何数据复制过程，也无需物理移动磁盘或阵列 LUN 。

关于此任务

- 由于在聚合重新定位操作期间会以编程方式验证卷计数限制，因此无需手动检查此限制。

  如果卷数超过支持的限制，则聚合重新定位操作将失败，并显示相关的错误消息。

- 在源节点或目标节点上正在执行系统级操作时，不应启动聚合重新定位；同样，也不应在聚合重新定位期间启动这些操作。

  这些操作可以包括以下内容：

  - 接管
  - 交还
  - 关闭
  - 另一个聚合重新定位操作
  - 磁盘所有权更改
  - 聚合或卷配置操作
  - 存储控制器更换
  - ONTAP 升级
  - ONTAP 还原
- 如果您使用的是 MetroCluster 配置，则不应在执行灾难恢复操作（ *switchover* ， *healing* _ 或 _*switchback* ）时启动聚合重新定位。
- 如果您拥有 MetroCluster 配置并在已切换的聚合上启动聚合重新定位，则此操作可能会失败，因为它超过 DR 配对节点的卷限制计数。
- 您不应对已损坏或正在进行维护的聚合启动聚合重新定位。
- 在启动聚合重新定位之前，应保存源节点和目标节点上的所有核心转储。

步骤

1. 查看节点上的聚合以确认要移动的聚合，并确保它们处于联机状态良好：

   `storage aggregate show -node` *`source-nod`*

   以下命令显示了集群中四个节点上的六个聚合。所有聚合均已联机。Node1 和 Node3 形成一个 HA 对，Node2 和 Node4 形成一个 HA 对。

```
cluster::> storage aggregate show
Aggregate     Size Available Used% State   #Vols  Nodes  RAID Status
--------- -------- --------- ----- ------- ------ ------ -----------
aggr_0     239.0GB   11.13GB   95% online       1 node1  raid_dp,
                                                         normal
aggr_1     239.0GB   11.13GB   95% online       1 node1  raid_dp,
                                                         normal
aggr_2     239.0GB   11.13GB   95% online       1 node2  raid_dp,
                                                         normal
aggr_3     239.0GB   11.13GB   95% online       1 node2  raid_dp,
                                                         normal
aggr_4     239.0GB   238.9GB    0% online       5 node3  raid_dp,
                                                         normal
aggr_5     239.0GB   239.0GB    0% online       4 node4  raid_dp,
                                                         normal
6 entries were displayed.
```

2. 使用问题描述命令启动聚合重新定位：

   s存储聚合重新定位 `start -aggregate-list` *`aggregate-1`* ， *`aggregate-2`* … `-node source-node -destination` *`destination-nod`*

   以下命令将聚合 aggr_1 和 aggr_2 从 Node1 移动到 Node3 。Node3 是 Node1 的 HA 配对节点。聚合只能在 HA 对中移动。

```
cluster::> storage aggregate relocation start -aggregate-list aggr_1,
aggr_2 -node node1 -destination node3
Run the storage aggregate relocation show command to check relocation
status.
node1::storage aggregate>
```

3. 使用 `storage aggregate relocation show` 命令监控聚合重新定位的进度：

   `storage aggregate relocation show -node` *`source-nod`*

   以下命令显示要移动到 node3 的聚合的进度：

```
cluster::> storage aggregate relocation show -node node1
Source Aggregate   Destination    Relocation Status
------ ----------- -------------  ------------------------
node1
       aggr_1      node3          In progress, module: wafl
       aggr_2      node3          Not attempted yet
2 entries were displayed.
node1::storage aggregate>
```

重新定位完成后，此命令的输出将显示每个聚合的重新定位状态为 `Done` 。

# 用于聚合重新定位的命令

有特定的 ONTAP 命令可用于在 HA 对中重新定位聚合所有权。

| 如果您要 ... | 使用此命令 ... |
| --- | --- |
| 启动聚合重新定位过程 | s存储聚合重新定位开始 |
| 监控聚合重新定位过程 | s存储聚合重新定位显示 |

"ONTAP 9 命令"

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